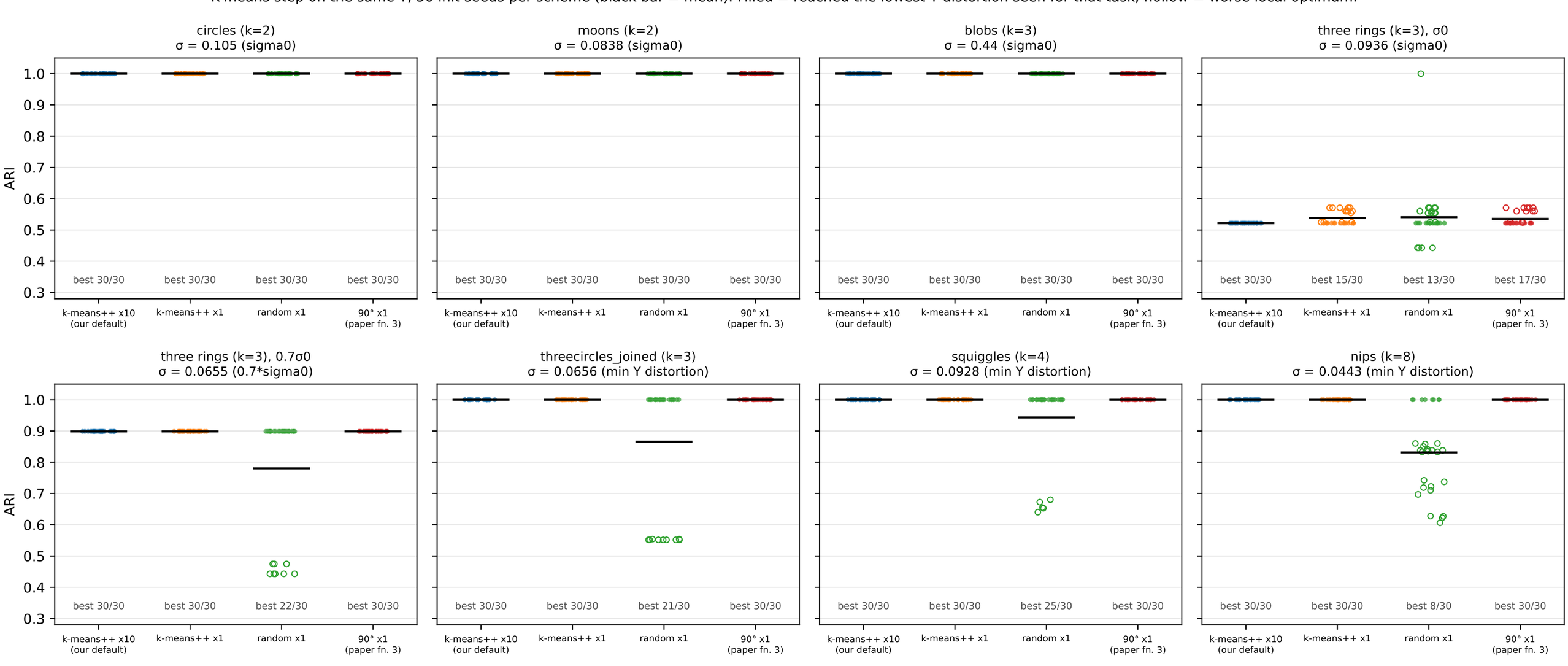
K-means step on the same Y, 30 init seeds per scheme (black bar = mean). Filled = reached the lowest Y distortion seen for that task; hollow = worse local optimum.
circles (k=2)
σ = 0.105 (sigma0)
moons (k=2)
σ = 0.0838 (sigma0)
blobs (k=3)
σ = 0.44 (sigma0)
three rings (k=3), σ0
σ = 0.0936 (sigma0)
three rings (k=3), 0.7σ0
σ = 0.0655 (0.7*sigma0)
threecircles_joined (k=3)
σ = 0.0656 (min Y distortion)
squiggles (k=4)
σ = 0.0928 (min Y distortion)
nips (k=8)
σ = 0.0443 (min Y distortion)
ARI
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
best 30/30
best 15/30
best 13/30
best 17/30
best 22/30
best 21/30
best 25/30
best 8/30
k-means++ x10 (our default)
k-means++ x1
random x1
90° x1 (paper fn. 3)